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A model to measure self-assessed proficiency in electronic medical records: Validation using maturity survey data from Canadian community-based physicians

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ABSTRACT

Objective: Adoption of electronic medical records (EMRs) does not necessarily translate to proficiency –referred to here as EMR maturity. To realize the full benefit of wide scale EMR adoption, the focus must shift from adoption to advancing mature use. This calls for validated assessment models so that researchers, health system planners and digital health developers can better understand what contributes to maturity among physicians. This research aims to validate a measurement model for self-assessed EMR maturity among community-based physicians.

Methods: As part of an Ontario government-funded EMR adoption program, the EMR Maturity Model for community-based practices was adapted from a hospital-based EMR maturity model. A survey instrument was developed on the foundation of the new model and revised by experts and stakeholders. Content validity, face validity and user acceptance were established before survey administration. Internal consistency and construct validity of the model were tested after survey data were collected. Finally, physicians' comments collected via the survey were qualitatively analyzed to provide additional insights that can be applied to refinement of the model and survey.

Results: As of August 1, 2019, 1588 physicians completed the survey. Ordinal alpha tests for reliability and content validity yielded an alpha value of 0.86 across all key measures specifically associated with maturity. Among most of these, there was a pattern of weak to moderate significant ($p < .0001$) positive Spearman inter-correlations. One factor was extracted for items measuring dimensions of maturity and all factor loadings of the key measures were greater than 0.40. The fit of the one-factor model was moderately adequate. This indicates the model is valid and reliable, with consistency across key measures for measuring one factor: maturity.

Conclusions: This is the first known validated model published in English that measures EMR maturity among community-based physicians. While the model is shown to be valid and reliable statistically and qualitative analysis supports this, there is room for improvement. Both the statistical analysis and portions of the qualitative analysis suggest areas of exploration to strengthen the model and survey. Future efforts will include refining the survey to improve user interface and accrue further data, as the sample to date is insufficient for generalizability.

1. Introduction

Electronic medical records or EMRs—systems that "organize and store medical records electronically" [1](p125)—have great potential to improve patient care through better practice management, information management, and diagnosis and treatment support [2]. At the broader healthcare system level, EMR use in community-based practices has shown benefits in population assessment and monitoring; chronic disease prevention and management; and reductions in costs, medical errors, adverse events, and duplicate tests [3]. In Canada, EMR adoption among community-based primary care or specialist physicians (for example, cardiologists, ophthalmologists, rheumatologists) has been supported by provincial government incentives [4], so that as of 2018 85 % of primary care physicians employed EMRs in their practice. However, adoption does not necessarily translate to proficiency, and the full benefit of wide scale EMR adoption will only be realized when

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Level	Criteria	Capabilities
5	INTEGRATE	Use of portals, hubs, attachment to provincial e-health platforms sharing data from the EMR.
4	POPULATION DATA USE	Dashboarding of whole populations, acting upon the whole, performing population analysis at the practice level.
3	LOOK AHEAD/ PREDICT	Reminders and alerts are used at the point of care. Searches are done regularly and scheduled for review.
2	EARLY DATA USE	Acting upon the output of episodic searches, quick entry tools, forms, calculators, etc.
1	ENTER DATA	Documentation occurs electronically. Progress notes, forms, and other documents are entered into the EMR.
0	PAPER	Processes are primarily paper-based.

Stage	HIMSS Analytics® EMRAM EMR Adoption Model Cumulative Capabilities
7	Complete EMR; External HIE; Data Analytics, Governance, Disaster Recovery, Privacy and Security
6	Technology Enabled Medication, Blood Products, and Human Milk Administration; Risk Reporting; Full clinical decision support (CDS)
5	Physician documentation using structured templates; Intrusion/Device Protection
4	Computerized Practitioner Order Entry (CPOE) with CDS; Nursing and Allied Health Documentation; Basic Business Continuity
3	Nursing and Allied Health Documentation; Electronic Medication Administration Record (eMAR); Role-Based Security
2	Clinical data repository (CDR); Internal Interoperability: Basic Security
1	Ancillaries - Laboratory, Pharmacy, and Radiology/Cardiology information systems; PACS; Digital non-DICOM image management
0	All three ancillaries not installed

Fig. 1. EMR Maturity Model and HIMSS EMRAM model [27].

physicians and their staff, with necessary supports, can become skilled users [6].

To improve, users need the means to measure their baseline proficiency – and track it over time as their skills mature. This is ideally accomplished with a validated measure: a framework or model, whose validity is confirmed via a questionnaire to collect data. Validation both confirms the model measures what it is supposed to measure, and gives confidence that the results will provide useful insights for respondents and researchers alike. Additionally, a rigorous measure of EMR maturity is of use in understanding how a discrete intervention – for example, integrating patient health data from other sources to the primary care EMR – helps physicians (and by extension, their patients) derive benefit from EMRs.

There is a paucity of research on EMR maturity measurement models; even those that exist are often not validated, which limits their utility [7]. Among validated EMR related models, we found that most of them focused on (1) behavioural factors surrounding EMR adoption, acceptance, intended use, or use [1,8–16] (2) user experiences such as EMR usability, quality, impact, training and support, usefulness, and satisfaction [4,17–25]. This gap in valid standard measures for EMR use was the impetus for a scoping review [26] where the authors found that almost all studies took a similar approach to studying EMR use via specific functions, under broad categories such as decision support and care management; yet this common ground has not led to a standardized measurement model. Even for the few well-known measures, e.g., HIMSS EMRAM [27] which assesses EMR users' function-based maturity against stages of EMR adoption and utilization, no evidence could be found on validity.

OntarioMD, the provincial government-funded delivery partner for EMR adoption among community-based physicians in Ontario, responded to this gap by developing the EMR Maturity Model (EMM). We developed a survey based on the model –the EMR Progress Assessment (EPA) - that could be used as a self-assessment tool by physicians. The EMM was originally used to benchmark milestones for physicians participating in the EMR adoption program funded by the province of Ontario, to help physicians offset the costs of acquiring an EMR for their practice. Once a critical mass of EMR adopters was achieved, the program came to an end and the purpose of the model and survey shifted to understanding how well EMR adopters were advancing in their use of applications and data.

While content validity, face validity and user acceptance were established for the model before the EPA was used in the field, statistical validity and reliability tests were not possible until sufficient data had been collected. Having accrued sufficient data, and as we prepare to update the EPA, we identified the opportunity to establish the validity of the model through statistical and additional qualitative tests. This allows us to both report on progress and identify ways to refine the model and survey. For the purposes of this paper, we restrict our analysis to data from primary care physicians not only because they comprise by far the majority of EPA respondents, but also because we recognize that the EPA cannot be answered in the same way by primary care and specialist physicians (e.g., the applicability of questions on prevention and screening for specialist care). The EPA currently employs some branching logic for specialists, and we are revisiting how to better accommodate these differences as we update it.

This paper describes the approach used in this study to validate the EMM as a model to measure EMR proficiency among community-based physicians, and inform the future improvement of the model and the survey based on it. A note on terminology: for the purposes of describing the method, we use the term *key measures* to refer to the EMR functionality items in the EPA, as depicted in Fig. 2, and *maturity level* to refer to the numbered levels on the EMR Maturity Model. In that sense, the items in the EPA are key measures, and they are measured by maturity levels on the EMM.

2. Material and methods

The study design was informed by an extensive literature review where the process of model validation was thoroughly articulated. Our validation process focused on three aspects of validity: *content* and *face*, through expert assessment and reliability testing (Cronbach's Alpha); and *construct* validity, through statistical tests. In the absence of comparable validated measures, we determined the optional test of *criterion* validity to be outside scope, especially as we had conducted the more important confirmatory tests of construct validity.

2.1. Model development

The EMM is an innovative model for measuring, understanding, and enhancing EMR use among community-based physicians. It was initially developed by two physicians and one eHealth expert by tailoring the eight-stage hospital-based HIMSS EMRAM [27] to community-based clinical practice. For example, high level HIMSS functions such as information sharing between hospital departments – irrelevant to the primary care setting – were replaced with integrated data access to regional and provincial resources such as portals and hubs. Fig. 1 compares the EMM and HIMSS EMRAM. The EMM was later revised in both structure and language based on feedback from members of the OntarioMD Peer Leader Program: community-based practitioners with advanced knowledge and experience of EMR-based clinical workflow who advise their colleagues on optimal EMR use. The EMM provides a detailed maturity roadmap that measures physicians' EMR capabilities on a six-point scale. The scale starts with no EMR use at all – possible in community-based practices (i.e., Level 0), and moves from basic use

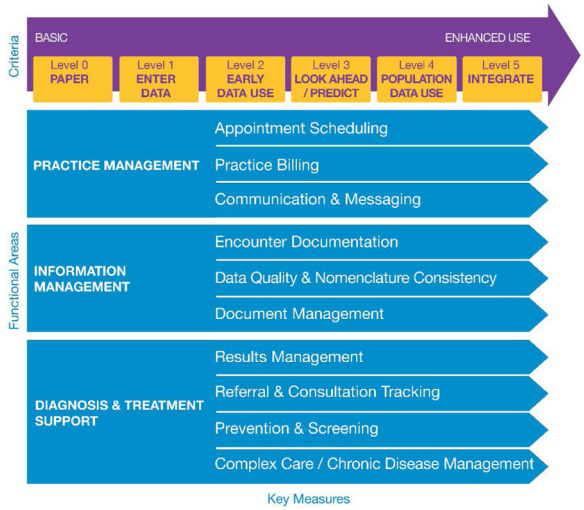

Fig. 2. Structure of EPA Clinical Questions.

(i.e., Level 1 and 2) to extended use (i.e. Level 3–5). Each maturity level corresponds to different EMR capabilities and builds upon the preceding level.

2.2. Survey development

A survey instrument based on the EMM was developed by a physician-led group and improved over time to become the current EPA. Questions were designed to elicit information about the respondent's use of EMR functionality in the context of their care delivery. For each EPA key measure – activities or processes such as appointment scheduling, prevention and screening, document management, data quality – respondents were given descriptive statements corresponding to each EMM level, but tailored to activities associated with that item, and asked to identify the level that best describes their use in that key measure. (Some non-clinical questions were added on satisfaction and user experience; outside this paper's focus).

In the clinical questions of the EPA, EMR maturity is assessed on three functional areas across 10 key measures (Fig. 2) reflecting aspects of clinical work most impacted by EMRs, and most pertinent for achievement of outcomes in efficiency, effectiveness, and patient-centred care. They are loosely aligned with the EMR features identified in other studies [8,26,28–32].

The complete EPA comprises three sections and seventy-five questions. Table 1 shows the key measures covered by the EPA; Table 2, an example of how maturity levels are described for a specific measure, in this case Appointment Scheduling. The first section collects demographic information. The second and core section is composed of 11 subsets of three clinical questions. Each subset matches one of the 10 key measures except for "Referral and Consultation Tracking", which branches separately for primary care providers and specialists. In each subset (Fig. 3), physicians select their perceived current ("Where I am now") and desired level ("Where I want to be next") of EMR maturity and leave comments or questions in a designated open field, if necessary. Those who selected "0" for their current maturity level are required to choose a reason for still employing a paper-based system. The third section focuses on physicians' appraisals of their EMR products as well as services and support received from OntarioMD.

2.3. Content validity and user acceptance

The first step in assuring the EMM's validity was engagement with subject matter experts. It was developed by a physician and peer-reviewed for relevance prior to development of the EPA survey tool. Content validity (how well the model reflects what it is intended to measure) was assessed by a professionally diverse group comprising a practice manager, nurse practitioner, practice advisor, practice enhancement consultant, and three physicians, who reviewed the draft EPA and confirmed key measures appropriately reflected the construct of maturity. Twelve peer leaders also completed the EPA for user acceptance testing.

These early content validity assessments informed EPA revisions, including reducing the number of key measures and functional areas, and changing language to reflect regional differences in digital integration initiatives. Based on user acceptance testing, further revisions included refining wording and reducing redundancy in the non-clinical EPA questions.

Table 1
Content of EPA.

Section	Subsection	No. of Questions	Subtotal
1.About You	Your Contact Details	11	24
	Your Practice Settings	6	
	Your EMR Solution	7	
2.Adoption Progress & Ease-Of-Use	● Practice Management		33
	Appointment Scheduling	3	
	Practice Billing	3	
	Communication & Messaging	3	
	● Information Management		
	Encounter Documentation	3	
	Data Quality & Nomenclature Consistency	3	
	Document Management	3	
	● Diagnosis & Treatment Support		
	Results Management	3	
	Referral & Consultation Tracking (Primary Care)	3	
	Referral & Consultation Tracking (Specialist)	3	
	Prevention & Screening	3	
	Complex Care/Chronic Disease Management	3	
3.Overall Experience	Practice Impact	2	18
	Service Quality	9	
	OntarioMD Support Experience	3	
	EMR Product and EMR Vendor Satisfaction	4	
		TOTAL	**75**

2.4. Field validation

2.4.1. Survey administration

The EPA was first administered to community-based physicians in Ontario in July 2016, in conjunction with the Ontario government-funded EMR adoption program. While incentives were in effect, OntarioMD practice advisors actively contacted program enrollees with instructions on how to access and complete the EPA online, and reminded them that completion was an eligibility requirement for the full incentive. Community-based physicians who did not participate in the EMR adoption program were welcome to complete the EPA without incentives. All physicians were encouraged to repeat their assessment at intervals to track their progress. Because physicians were not recruited randomly and most were incentivized for participation, findings are not generalizable to the general population of Ontario community-based physicians.

The EPA continues to be accessible to all community-based physicians in Ontario via the secure online portal at ontariomd.ca. At login, physicians are provided a link to the informed consent statement (Appendix A). During survey completion, physicians can use the "Finish Later" button to save their results and the "Resume" button to return to the page where they left off. Upon completion, they may view graphics

Table 2
Characteristics of Sample Physicians.

Role	n (%)	Year of Birth		Practice Type	n (%)	EMR Solution	n (%)
Family Physician	1071 (67.44)	Range	1933 – 1995	Primary Care Group Practice	881 (55.48)	PS Suite	552 (34.76)
Specialist	517 (32.56)	Mean	1968	Specialist Group Practice	240 (15.11)	Accuro	349 (21.98)
		Std	12.1 yrs	Sole Practitioner	467 (29.41)	Oscar	332 (20.91)
						Clinic Information System	96 (6.04)
						Other	259 (16.31)

depicting their results and progress.

2.4.2. Statistical tests of validity and reliability

Internal consistency of key measures and construct validity were assessed to test the reliability and validity of the EMM. One factor – self-assessed current EMR maturity level – was prioritized for validity assessment because it was the key survey outcome (and more reliable than desired maturity level).

Statistical tests commonly used for validity and reliability included Cronbach's alpha coefficient [33] to measure internal consistency, with the acceptable threshold value being greater than 0.7 [23]. Internal consistency was also analyzed through inter-item Spearman's rank-order correlation tests with adjusted *p* values using the Holms Method. Because the maturity level data was ordinal, we employed polychoric rather than Pearson's correlation tests for confirmatory factor analysis to establish construct validity [34]. The minimal residual method was used based on a polychoric correlation [35] matrix with equamax rotation to test one, two or three factors or item groupings. Parallel analysis for polychoric correlations was conducted, and a scree plot of observed and simulated data was generated. The criteria used to confirm the optimal number of factors extracted included (1) factor loadings > 0.40; (2) the variance explained by the factors; (3) eigenvalues > 1; (4) scree plot [33]. The model fit adequacy was assessed by Tucker Lewis Index (TLI), Root Mean Square Residual (RMSR), and Root Mean Square Error of Approximation (RMSEA). R (version 3.5.3) with packages such as *tidyverse*, *RcmdrMisc*, and *psych* were used in the statistical validation analysis.

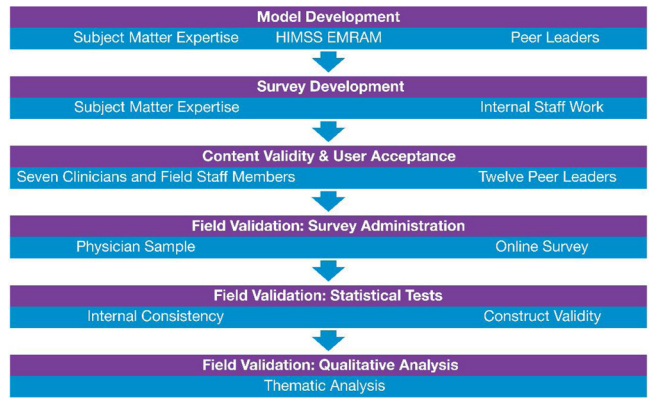

Fig. 4. Study Design Process Map.

2.4.3. Qualitative analysis

To maximize the value derived from EPA data, we conducted open coding of text responses to unearth insights of value for both the validation process and the planned update of the EPA. Respondents were offered the option to elaborate in text boxes following each key measure question; a small percentage availed themselves of this option. Comments associated with each key measure were imported into NVivo 12 Plus (trial version) and coded by theme. No *a priori* codes were used to guide the analysis so that themes could naturally emerge from the texts.

25.		Appointment Scheduling*	Where I am now	Where I want to be next
	0	I book my appointments outside of the EMR, on paper.	□	□
	1	I book my appointments using the EMR, but perform no analysis on wait times, etc.	□	□
	2	I use scheduling templates to ease workflow and create openings for urgent care (i.e., same day visits).	□	□
	3	I have implemented an advanced access schedule, with or without the use of an online scheduling system, with consideration of same day/next day appointments	□	□
	4	I have implemented an online booking system and perform analysis of supply and demand of clinicians' time, mapping to the availability of appointments. Advanced access scheduling is implemented with ongoing local analysis	□	□
	5	I have implemented an integrated e-booking system with advanced access and perform analysis of third next available and same day/next day appointments. Analyses are dashboarded and reported for benchmarking purposes	□	□
26.		**Do you have any additional comments or questions? (optional)**		
27.		**Which of the following best describes why you are still using a paper-based system for appointment scheduling?** * *(Please respond only if you selected Level 0 in the "Where I am now" column for this function)* □ I do not use the EMR for this function currently, but plan to do so within the year. □ I have tried it, but do not typically use the EMR for this function. □ I need training on how to use the EMR for this function. □ I do not have this function currently available on my EMR □ I have no plans to use the EMR for this function. □ I don't know. □ Other, please specify:		

Fig. 3. A Sample Screenshot of EPA Clinical Questions.

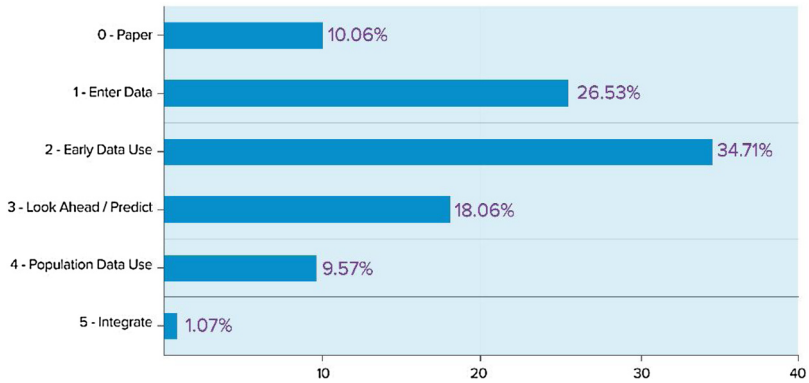

Fig. 5. EMR Maturity Aggregated Across Key Measures.

An overview of this study design is shown in Fig. 4.

3. Results

3.1. Physician sample

Between July 2016 and July 2019 inclusive, 1983 EPAs were collected. After removing duplicates (i.e., physicians who took more than one EPA) and ineligible responses (e.g., those with invalid College of Physicians and Surgeons of Ontario numbers), 1588 unique records remained for analysis; 1485 (93.5 %) of whom had participated in the EMR adoption program. This represents a response rate of 12.1 % of the 12,242 who participated in the incentive program. Although, a sample size of 1588 physicians provides sufficient statistical power to conduct inferential tests, it should be noted that selection bias (i.e., most physician participants were incentivized as part of the EMR adoption program) precludes generalization to the general population of community-based physicians in Ontario. Table 2 shows the characteristics of the sample physicians. More than half were at Level 1 or 2 for current EMR maturity (Fig. 5).

3.2. Statistical tests

The ordinal alpha tests for polychoric correlations for the key measures yielded an alpha value of 0.86 across all the key measures, demonstrating these measures are mutually consistent and interrelated (i.e., the alpha falls between .70 and .95 [36]). Tests also showed evidence of little change in alpha (0.84 or 0.85 for each key measure) if subsequent key measures were dropped, indicating reliability in that each key measure pertains to the same thing – i.e., maturity. For most key measures, there was a weak to moderate significant ($p < .0001$) positive Spearman correlation within the maturity levels (Appendix C). Table 3 shows the results of factor analysis for both the one-factor and two-factor models.

One factor was extracted for the maturity levels and all the factor loadings of the key measures were greater than 0.40. Fig. 6 shows the standardized loadings for each key measure, in order of the strength of their polychoric correlation with the derived construct of maturity. For this first factor, some key measures – notably Prevention & Screening, and Complex Care / Chronic Disease Management - showed particularly strong correlation with maturity. Some key measures had loadings greater than 0.3 on the second factor. The eigenvalue was 3.90 (> 1) for the one-factor model and 0.64 (< 1) for the two-factor model. The scree plot with eigenvalues for the EPA data compared with random simulated data also indicated a one-factor solution (Fig. 7). These results support validation as they indicate there is agreement and consistency across key measures, and these are measures of one factor alone: maturity.

The fit of the one-factor model was moderately adequate (RMSR = 0.06, RMSEA = 0.129 with 90 % CI [0.122, 0.136], TLI = 0.79), confirming that the key measures do indeed measure maturity. A two-factor model yielded a marginally better fit, however the eigenvalue for the second extracted factor was less than one. Thus, the two-factor solution was rejected, and the one-factor solution retained. Dropping the key measures with lower factor loadings provided no benefit and even weakened the model fit. Dropping one of the key measures only caused slight improvements.

3.3. Qualitative analysis

In the EPA, respondents have the option to elaborate on their responses via open text questions. Across all key measures, between 2 % and 14 % of respondents did so; therefore, comments cannot be viewed as generalizable, but deliver context to user experience that may be helpful as we revise the EPA. Open coding was conducted of these text comments to identify emergent themes. This analysis identified five scenarios where physicians may have selected lower maturity levels unrelated to EMR maturity: the more advanced levels were (1) not applicable given the nature of their practice or patient population; (2) not practical due to technical difficulties, financial challenges, or external constraints; (3) not desirable due to low priority, malfunctions, or satisfaction with current use; (4) not familiar due to being out of the scope of responsibilities; (5) not universal due to only being carried out in some aspects of the clinical practice or health conditions under care.

4. Discussion

Using the validation approach outlined in this paper, we can confirm that the EMR Maturity Model is both reliable and valid in measuring EMR maturity among community-based physicians. There is, however, room for improvement, and as we work towards the next iteration of the EPA. This study suggests several helpful insights.

For internal consistency, alpha tests yielded high scale reliability across all key measures, indicating that if physicians were to repeat the EPA several times during a short period, they would select the same maturity levels most of the time. Little changes in the alpha were observed when each key measure was dropped, suggesting the possible presence of multicollinearity and thus redundancy. This is an important consideration in revising the EPA; if we can eliminate redundancy and shorten the survey, it may be to the benefit of our response rate. The key measures were not highly correlated in Spearman's tests, probably because the data were ordered and non-numeric, or physicians' maturity levels were too homogenous as shown in Fig. 4. The correlations were still significant even when the effect size was low because the sample size was relatively large.

For construct validity, all the key measures were strongly correlated with the one confirmed factor extracted from the factor analysis and each key measure provided an estimate for it. However, the model fit

Table 3
Standardized Loadings (Pattern Matrix) Based Upon Correlation Matrix.

Key Measure	Factor 1	h^2 *	u^2 **	Factor 2	h^2 *	u^2 **	No. of Component
Appointment Scheduling	0.56	0.31	0.69	0.09	0.30	0.70	1
Communication & Messaging	0.63	0.40	0.60	0.31	0.45	0.55	1
Practice Billing	0.55	0.30	0.70	0.08	0.30	0.70	1
Encounter Documentation	0.71	0.51	0.49	0.15	0.50	0.50	1
Data Quality & Nomenclature Consistency	0.54	0.30	0.70	−0.12	0.34	0.66	1
Document Management	0.55	0.31	0.69	0.34	0.37	0.63	1
Results Management	0.57	0.32	0.68	0.47	0.49	0.51	1
Referral & Consultation Tracking	0.67	0.45	0.55	0.25	0.47	0.53	1
Prevention & Screening	0.70	0.48	0.52	−0.14	0.57	0.43	1
Complex Care/Chronic Disease Management	0.72	0.52	0.48	−0.29	0.74	0.26	1

	Factor 1	Factor 2
SS Loadings	3.90	0.64
Proportion Variance	0.39	0.06
Proportion Explained	0.86	0.14

* Squared factor loading, i.e., how much variance in the item is explained by the factor.
** 1-squared factor loading, i.e., how much variance in the item is not explained by the factor but undetermined causes.

was only moderately acceptable and there was some unexplained variance. A few key measures loaded relatively high on the second factor, suggesting the possibility of measuring another unknown construct and thus weakening the model fit.

Although our findings confirm the validity of the tool, results must be interpreted with caution in terms of their representativeness, given the low response rate and the potential for non-response bias. Similar to a convenience sample, more than 80 % of respondents were drawn from the pool of Ontario community-based physicians enrolled in the EMR adoption program. With EPA completion required for an incentive payment, there is a possibility some respondents completed to "tick the box" for that milestone. However, this concern is complicated by the reality that not all physicians in the program completed an EPA, and not all EPA completers were in the program – some completed even after the program ended. Notably, as participants in the program represented the majority of Ontario community-based physicians and were geographically distributed, a greater response rate from our pool has the potential to be representative.

Our findings may also be limited by the inherent weaknesses in self-reported assessments, i.e., potential bias from intention to meet certain expectations, involuntarily filtered memories [37,38], cognitive load, or available time [39]. Actions we could take to accumulate more evidence of high quality and wide breadth to support the validity of the EMM [7] include:

- First, comparing the survey results with other more objective measures for triangulation, e.g., standardized clinical indicators of quality care [2], EMR logs or audit trails [15,38–40], assessments of field observers [2,39], concurrent assessments of multiple physicians from the same practice [4].
- Second, as the findings of statistical validation indicate some redundancy, shortening the survey may encourage compliance and drive a higher response rate in future – especially important given the financial incentive program has ended, and we are seeking new ways to accrue data to the sample (for example, using opportunistic collection at conferences). We could further draw from other studies that, noting the tendency towards low survey response rates among clinicians, have explored ways to drive rates higher in this group. [41,42]
- Third, identifying new applications of this model for quality improvement programs could bolster evidence of its validity and preclude unnecessary allocation of resources to proliferation of further models [7]. In future analyses, it would be instructive to further consider demographic, socioeconomic, organizational, and infrastructural factors that may affect EMR maturity. This will be especially valuable should the measure's use be extended to other jurisdictions, where its value could be more widely demonstrated as a trustworthy measure.

5. Conclusion

Informed by best practices identified in the academic literature, this research validated a model to measure EMR maturity among community-based physicians. To the best of our knowledge, this is the first

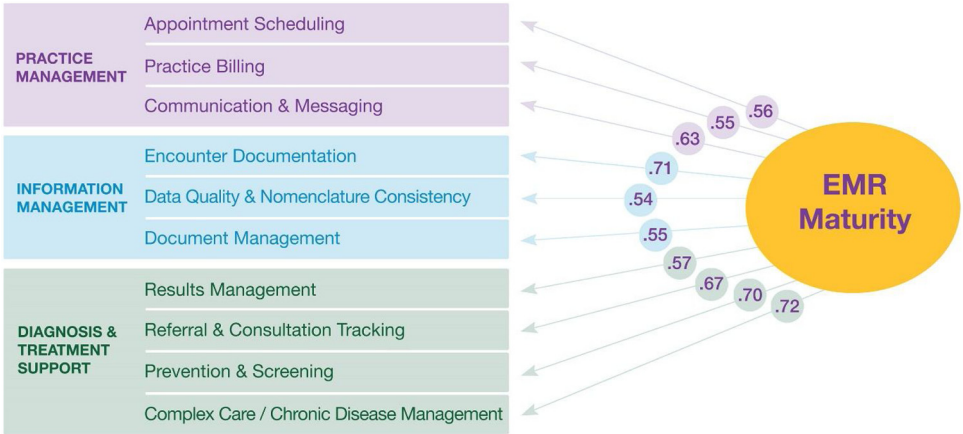

Fig. 6. Standardized loadings for each key measure (correlation with the construct of maturity).

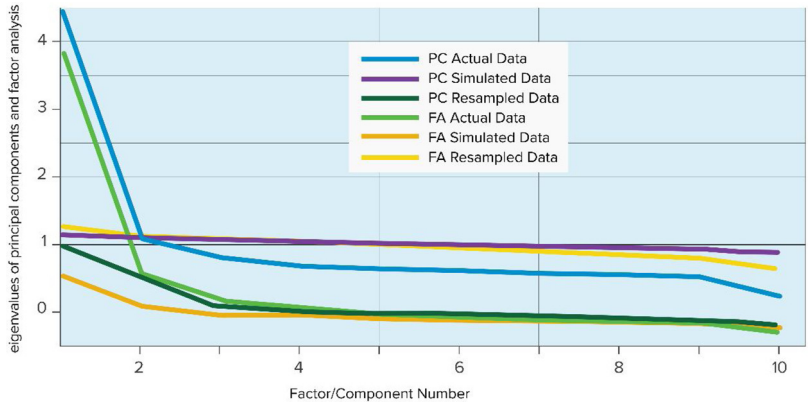

Fig. 7. Scree Plot Showing One-Factor Solution.

validated measure for this purpose published in English. With this measure, physicians' EMR capabilities can be better identified and corresponding supporting resources can be more effectively mobilized to move physicians up in the maturity scale. Ultimately, more EMR benefits can be realized in the improvement of healthcare quality and patient outcomes. Finally, though this analysis has been successful in establishing the validity of the EMM in relation to maturity, the model could be strengthened and we will explore options to do so.

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CRediT authorship contribution statement

Jing Chong: Conceptualization, Investigation, Formal analysis, Methodology, Writing - original draft. **Timothy Jason:** Formal analysis, Methodology, Data curation, Validation, Visualization, Writing - original draft, Writing - review & editing. **Mavis Jones:** Conceptualization, Supervision, Writing - original draft, Writing - review & editing. **Darren Larsen:** Conceptualization, Methodology, Resources.

Declaration of Competing Interest

None.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.ijmedinf.2020.104218.

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